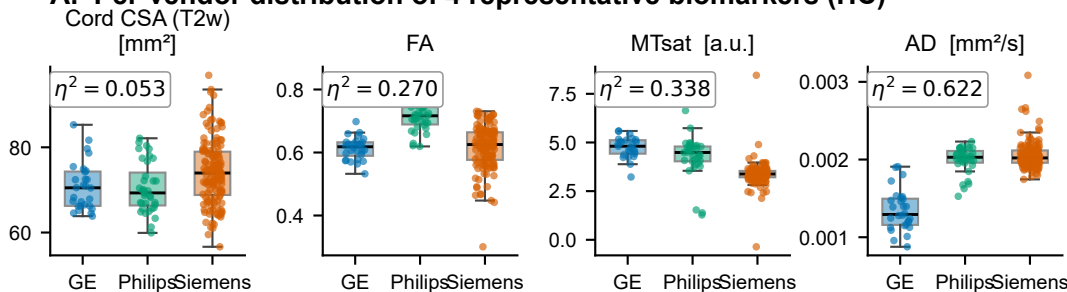


Baseline vendor effects in unharmonized spinal-cord qMRI biomarkers

A. Per-vendor distribution of 4 representative biomarkers (HC)



B. UMAP of standardized 8-biomarker vector

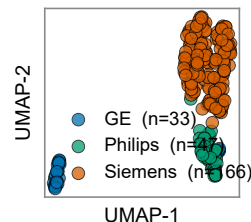
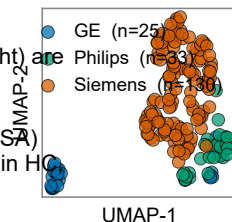
HC (N = 188)

ALL (N = 246)

Findings — baseline vendor effects

- Demographic covariates (age, sex, height, weight) are balanced across vendors (all $p > 0.1$, Table 1).

- Univariate vendor η^2 ranges from 0.05 (Cord CSA) to 0.62 (AD) — mean 0.28 across 8 biomarkers in HC, 0.30 in ALL.



- Multivariate PERMANOVA $R^2 = 0.315$ (HC, $p < 10^{-4}$)

C. Baseline vendor η^2 per biomarker with multivariate PERMANOVA R^2 as reference

about one third of the joint biomarker distance.

